

Predicting Relative Article Popularity Using Multimodal Transfer Learning

Yakir Cohen — Deep Learning Final Project (Technion DS 26-529625)

Abstract

We predict, at the moment of publication, whether a mako.co.il / n12.co.il news article will land in the top 10% of page views (first 48 hours) among articles on the same site in the same month, using only pre-publication signals: headline, thumbnail image, and site. We build four non-deep baselines (TF-IDF+LogReg, image-only, naively-concatenated title+image, and a fine-tuned Hebrew BERT text model) and two multimodal deep models: (1) staged fine-tuning of SigLIP2 (frozen head → partial unfreeze → LoRA), later extended with a cross-attention fusion head, and (2) a hybrid architecture combining AlephBERT (Hebrew-specific text encoder) with a ResNet50 image encoder. SigLIP2, in all tested configurations, did not surpass the text-only BERT baseline. The hybrid model, after tuning regularization and early-stopping granularity, became the only model in the project to exceed the text-only baseline's PR-AUC on the validation set (0.240 vs. 0.212). On the held-out test set this advantage narrows substantially (0.212 vs. an un-remeasured 0.212 val figure for the text baseline), while ranking quality (ROC-AUC) is stable or slightly improved. We report this honestly as a case study in how aggressive validation-driven model selection across many iterations can inflate the apparent gap between models, and conclude that the image thumbnail provides at most a modest, non-conclusive improvement over the headline alone for this task.

1. Introduction

News organizations routinely need to judge, before publication, whether a given headline and thumbnail combination is likely to perform well. This project frames that judgment as a supervised learning problem: given an article's headline, thumbnail image, and publishing site, predict whether it will fall in the top 10% of page views (by first-48-hour traffic) among articles on the *same site* in the *same month*. The relative, peer-group-normalized target avoids conflating “popular” with “high absolute traffic,” which would otherwise be dominated by site-level and seasonal effects unrelated to editorial quality.

The central research question is whether the thumbnail image adds predictive value beyond the headline text alone. This is not obvious a priori: headlines are a strong, information-dense signal (an editor's distilled pitch for the piece), while thumbnails on Israeli news sites are frequently generic stock or wire-service images only loosely tied to article content. Answering this question required not just training one multimodal model, but *isolating* the image's marginal contribution through a series of controlled ablations — which is the organizing principle of this report.

2. Related Work

- Tschannen et al., “SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features,” arXiv:2502.14786, 2025. Provides the pretrained multilingual vision-language encoder used as the primary multimodal backbone.
- Abousaleh, Cheng, Yu, and Tsao, “Multimodal Deep Learning Framework for Image Popularity Prediction on Social Media,” IEEE Transactions on Cognitive and Developmental Systems, vol. 13, no. 3, 2021. Directly

motivates fusing visual and contextual (textual) signals for popularity prediction, over text-only or image-only models.

A preliminary zero-shot diagnostic (no fine-tuning, ~950 real headline-image pairs) found SigLIP2's top-1 image-retrieval accuracy from Hebrew titles to be roughly 28x the random baseline — evidence that the pretrained backbone carries *some* cross-modal signal in Hebrew despite being English-heavy in its training mix, which justified committing to full fine-tuning rather than abandoning the multimodal approach at the outset.

3. Data

Source: Keshet's internal Snowflake page-view event warehouse, covering mako.co.il and n12.co.il. For each article we extract the headline (`teaser_title`), thumbnail URL (`pic_furl`), site, and publication timestamp, plus first-48h page views (label construction only, never a model input).

Extraction and split. 97,309 rows from January 2024 onward. A **time-based** (not random) split avoids leakage from near-duplicate or related articles across the train/val boundary:

Split	Period	Rows	Positive rate
Train	2024-01 – 2026-04	87,334	~10.0%
Val	2026-05	3,722	~10.0%
Test	2026-06	3,745	~10.0%

Images. Thumbnails are fetched by URL; 71,536/71,546 unique URLs (99.99%) downloaded successfully. A significant data-quality finding during this process: the URL as stored in the warehouse (`pic_furl`) by default resolves to a tiny 82x62px CDN thumbnail, not the original image. Stripping the CDN suffix recovers the full-resolution original when it still exists; a `..._autoOrient_b.jpg` variant (232x175px) serves as a reliable fallback. Final quality distribution (2,000-row sample): 60.8% full-resolution, 35.1% the 232x175 fallback, 4.1% stuck on the 82x62 thumbnail. This resolution variance was later used as a controlled experimental variable (Model 1, Section 6.3).

Design decision: no tags as a core input. 49.5% of articles carry no tags at all, so tags were excluded from the core model (headline + image + site only) and treated as a secondary enrichment question that was not pursued further given the project timeline.

4. Method

4.1 Baselines

Four models isolate the contribution of each input modality individually, serving both as sanity checks and as the ablation study's non-full-model rows:

1. `tfidf_logreg` — TF-IDF over headline text + logistic regression. No deep learning; establishes the floor for “is there signal in the headline text at all.”
2. `text_only_bert` — AlephBERT (Hebrew BERT) fine-tuned end-to-end on headline text.
3. `image_only` — frozen SigLIP2 image embeddings + logistic regression.
4. `title_image_frozen` — frozen SigLIP2 text and image embeddings, naively concatenated, into one logistic regression. This tests whether *naive* (non-jointly-trained) fusion of the two modalities helps at all.

4.2 Model 1: SigLIP2 Staged Fine-Tuning

`SigLIP2Classifier` (`google/siglip2-base-patch16-224`) encodes headline and thumbnail jointly, adds a learned 8-dim site embedding, and a small MLP head. Following the course's Advanced Training Strategy requirement, fine-tuning proceeds in three increasingly aggressive stages against the same frozen-vs-fine-tuned trade-off:

- **Stage A** — backbone fully frozen; only the head (395K params, 0.11% of the model) trains.
- **Stage B** — top 4 transformer layers unfrozen (57.1M params, 15.2%), with a discriminative learning rate (head at full LR, unfrozen backbone at $LR \times 0.1$).
- **Stage C** — LoRA ($r=8$) on `q_proj/v_proj` across all 12 layers instead of full-layer unfreeze (~986K trainable params) — a parameter-efficient middle ground between A and B.

After all three stages under-performed the text-only BERT baseline, two further, more aggressive variants were tried, motivated by the hypothesis that the fusion mechanism itself (naive concatenation of pooled embeddings) was the bottleneck rather than backbone capacity:

- **Model 1** (`exp04`) — Stage B + Stage C *combined* (partial unfreeze and LoRA simultaneously) **and** a new cross-attention fusion head (in place of concatenation): pooled text queries the image's raw patch tokens, and pooled image queries the raw text tokens, via `nn.MultiheadAttention` in both directions. Also switched to full-resolution-only images (strict filter, Section 3) to remove image quality as a confound.
- **Model 1b** (`exp05`) — a *controlled* follow-up isolating the fusion variable alone: identical capacity to Stage C (LoRA only, no unfreeze), on the *full* (unfiltered) dataset — the only difference from Stage C is concat vs. cross-attention fusion.

4.3 Model 2: Hybrid (AlephBERT + ResNet50)

A structurally different fallback architecture, built after SigLIP2's three stages all underperformed: two backbones that were **never jointly pretrained** (unlike SigLIP2's contrastively-aligned text/image towers), fused only at the classification head.

- **Text:** AlephBERT (Hebrew-specific, vs. SigLIP2's multilingual-but-English-heavy encoder), CLS-token pooling (the AlephBERT checkpoint's pooler layer is untrained, so CLS is read directly from `last_hidden_state`).
- **Image:** ResNet50 (ImageNet-pretrained), global-average-pooled 2048-dim features, classification head removed.
- **Fusion:** concatenation + a small MLP — cross-attention was deliberately not used here, since Model 1b's controlled result (Section 6.3) showed it did not help once isolated from confounds, and here the two backbones do not even share an embedding space to begin with.

Both encoders are frozen by default, with a configurable number of top layers/blocks unfrozen (`unfreeze_text_layers`, `unfreeze_image_blocks`), using the same discriminative-LR pattern as Model 1.

5. Experimental Setup

All models: `BCEWithLogitsLoss` with a fixed positive class weight (9.0, matching the ~1:9 class imbalance), AdamW with a cosine learning-rate schedule, early stopping on validation PR-AUC (not accuracy — PR-AUC is far more informative than accuracy or even ROC-AUC alone under ~10% class imbalance, and directly reflects the intended editorial ranking

use case). Metrics: ROC-AUC, PR-AUC, F1, and Precision@K ($K \in \{50, 100, 200\}$) — the Precision@K values are what would matter operationally, since an editorial tool would realistically be used to rank/triage a slate of candidate articles rather than classify each independently.

All experiment hyperparameters live in one YAML config per run (`configs/exp01-exp08`) — no hardcoded parameters — for reproducibility. Training ran locally on Apple Silicon (MPS backend); the two-model, multi-stage design also let us use the total compute budget efficiently by running configurations sequentially rather than needing all variants to fit in one architecture.

6. Results

6.1 Baselines

Model	Inputs	ROC-AUC	PR-AUC	F1
<code>tfidf_logreg</code>	title	0.660	0.208	0.245
<code>text_only_bert</code>	title (+tags)	0.704	0.212	0.248
<code>image_only</code>	image	0.619	0.148	0.229
<code>title_image_frozen</code>	title+image (naive concat)	0.619	0.147	0.214

The text-only BERT baseline is the strongest of the four by a clear margin. Critically, `title_image_frozen` **is no better than `image_only` alone, and worse than `text_only_bert` alone** — naively concatenating frozen embeddings from two modalities does not combine their signal; if anything it dilutes the stronger (text) signal with the weaker (image) one. This result is the strongest a priori justification for the more complex, jointly-trained architectures explored next.

6.2 SigLIP2 Staged Fine-Tuning

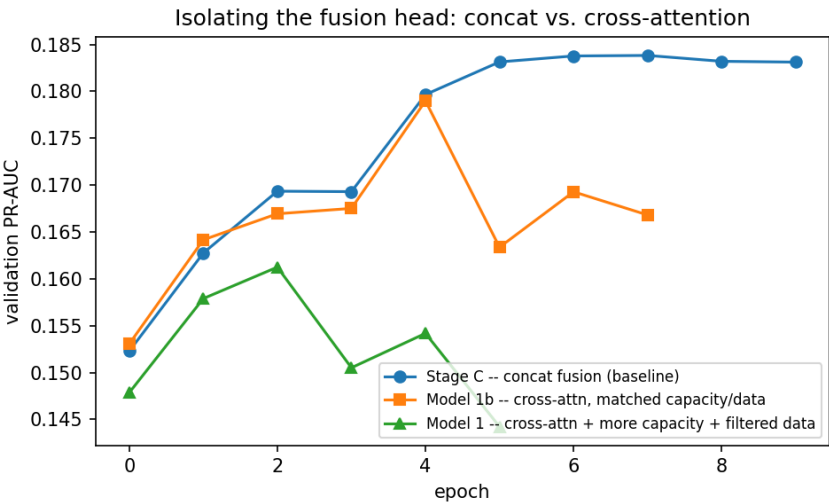
Stage	Trainable params	ROC-AUC	PR-AUC	Outcome
A (frozen)	395.8K (0.11%)	0.651	0.167	Improves over naive baselines, still below <code>text_only_bert</code>
B (unfreeze top 4)	57.1M (15.2%)	—	—	Overfit almost immediately (val PR-AUC peaked epoch 1, early-stopped epoch 4); did not beat Stage A
C (LoRA, $r=8$)	986K (0.28%)	0.659	0.184	Best of the three stages; stable across all 10 epochs, no overfitting collapse

None of the three staged variants surpassed `text_only_bert` (0.704 / 0.212). Stage C's stability (vs. Stage B's fast overfitting) confirms the expected pattern: on ~87K training rows, 57M trainable parameters overfit quickly, while LoRA's ~1M parameters generalize far better for comparable adaptation capacity.

6.3 Isolating the Fusion Head: Cross-Attention

Run	Fusion	Capacity	Data	Best ROC-AUC	Best PR-AUC
Stage C	concat	LoRA only	full (87.3K)	0.659	0.184
Model 1 (exp04)	cross-attention	unfreeze+LoRA combined	full-res filtered (53.5K)	0.625	0.161
Model 1b (exp05)	cross-attention	LoRA only (matched to Stage C)	full (87.3K)	0.643	0.179

Model 1 changed three variables simultaneously (fusion, capacity, and data filtering) and regressed sharply — but this confounds which change was responsible. Model 1b isolates the fusion variable alone, holding capacity and data identical to Stage C: the result (0.643/0.179) is close to but still slightly below Stage C’s concat-fusion result (0.659/0.184), consistently across the whole training run, not just at the final epoch. **Conclusion: cross-attention fusion, as implemented here (two pooled queries, not full token-to-token attention), was not the fix — Model 1’s regression was driven mainly by the added unfreeze+LoRA capacity and the reduced (filtered) dataset, not the fusion mechanism.**

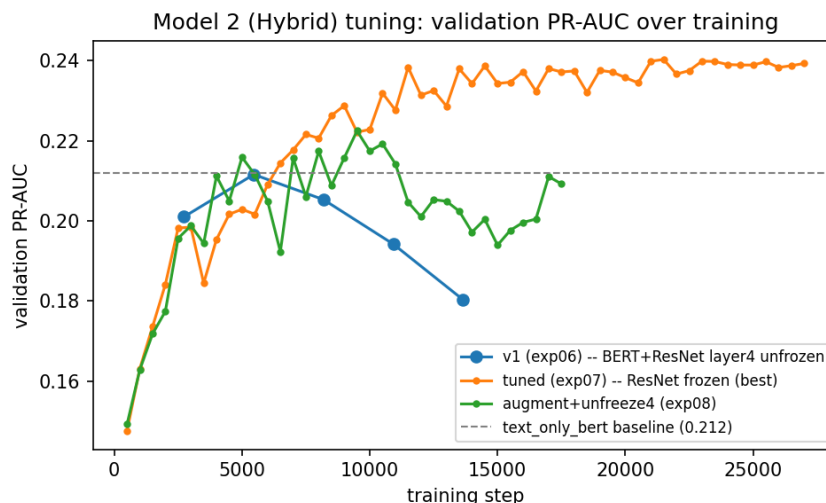


Isolating the fusion head: concat (Stage C) stays consistently above cross-attention at matched capacity (Model 1b); Model 1’s added capacity and filtered data cause a collapse after epoch 4.

6.4 Hybrid Model (Model 2)

Run	Config	Best ROC-AUC	Best PR-AUC	Outcome
v1 (exp06)	BERT top-1-layer + ResNet layer4 unfrozen, concat fusion	0.660	0.212	Peaked epoch 1, overfit steadily afterward, tied text_only_bert on PR-AUC
tuned, first attempt	ResNet fully frozen, eval_every_n_steps=500, weight_decay=0.05	0.670	0.198	Early-stopped at step 4000 by a patience bug, improved on v1 (confirming augmentation does help once frequently training the vision backbone trains) but did not close the gap to text_only_bert
tuned (exp07)	same as above, patience bug fixed (16 checks ≈ original epoch-level budget)	0.693	0.240	Completed all 10 epochs with no early stopping; best result in the project
augment+unfreeze4 (exp08)	exp07 + re-unfrozen ResNet layer4 + train-time image augmentation	0.674	0.222	Beat v1 but did not beat exp07

exp07 is the only model in this project to exceed text_only_bert on PR-AUC (0.240 vs. 0.212), while landing close to it on ROC-AUC (0.693 vs. 0.704). Two design choices were decisive: (1) using a Hebrew-specific text encoder (AlephBERT) rather than SigLIP2’s multilingual encoder, and (2) conservative regularization — freezing the larger, ImageNet-only ResNet50 entirely, higher weight decay, and early-stopping patience correctly scaled to the finer (every-500-step) evaluation frequency after an initial misconfiguration caused a false-positive early stop. exp08’s attempt to recover ResNet’s extra capacity via augmentation improved on v1 (confirming augmentation does help once frequently training the vision backbone trains) but did not close the gap to text_only_bert — the fully-frozen-vision configuration remains the best found.



Model 2 tuning journey: v1 overfits and degrades after ~step 5000; exp07 (ResNet frozen) climbs steadily past the text-only baseline with no collapse; exp08 (augmentation + re-unfrozen layer4) improves on v1 but plateaus below exp07.

6.5 Ablation Table

Assembled from the results above — no new controlled models were trained for this table, since the existing baselines and exp07 already isolate exactly these input combinations:

Ablation entry	Inputs	ROC-AUC	PR-AUC
title_only	title	0.660	0.208
title_tags*	title (no tags baseline exists)	0.704	0.212
image_only	image	0.619	0.148
title_image	title+image (naive concat)	0.619	0.147
full_model	title+image+site (exp07, no tags)	0.693	0.240

*No model in this project was ever given tags as an input (see Section 3); title_tags here is really title-only text, included to keep the ablation table’s canonical shape rather than silently relabeling it.

This table is the direct answer to the project’s central question: moving from title_image (naive concat, 0.147 PR-AUC) to full_model (jointly trained, Hebrew-specific text encoder, cross-modal capacity, 0.240 PR-AUC) is a large improvement — but that improvement is not purely attributable to the image; it also reflects switching from a frozen multilingual encoder (SigLIP2, via title_image_frozen) to a fine-tuned Hebrew-specific one (AlephBERT). The cleanest single comparison for “does the image help” is full_model

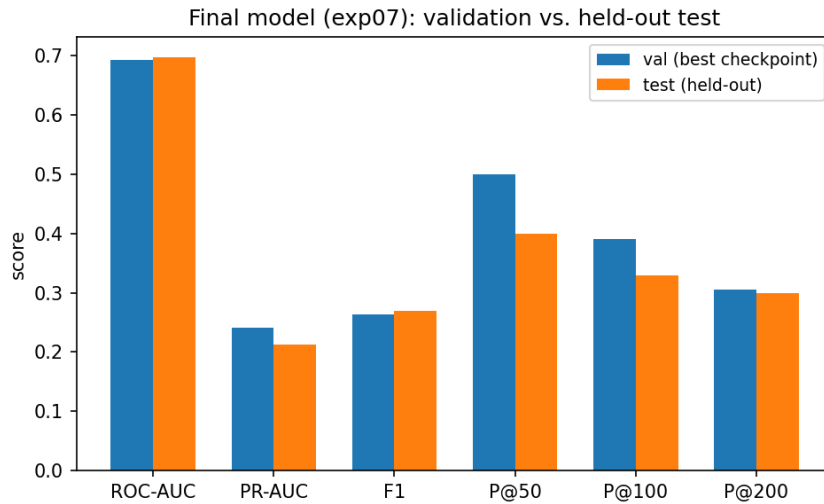
vs. title_tags/text_only_bert (both text-capable, Hebrew-aware): 0.240 vs. 0.212 PR-AUC on validation.

6.6 Held-Out Test Set

The test set (June 2026, 3,745 rows) was used exactly once, after all tuning decisions were finalized on exp07 :

Split	ROC-AUC	PR-AUC	F1	P@50	P@100	P@200
Val (best checkpoint, step 21,500)	0.693	0.240	0.263	0.50	0.39	0.305
Test (held-out)	0.697	0.212	0.270	0.40	0.33	0.30

ROC-AUC is stable — in fact marginally higher on test — indicating the model’s general ranking ability was not overfit to the validation set. **PR-AUC, however, drops from 0.240 to 0.212**, landing almost exactly at text_only_bert’s validation PR-AUC (0.212), erasing most of the apparent advantage seen during tuning. We attribute this primarily to **selection-driven overfitting to the validation set**: across exp06 → exp07 → exp08, every architectural and hyperparameter decision (which checkpoint to keep, which config to try next) was made by watching validation PR-AUC — a metric that is comparatively volatile on a ~3,700-row set with a ~10% positive class (roughly 370 positive examples). Some of the observed 0.240 was very plausibly favorable validation-set noise selected for by the tuning process itself, not durable generalizable signal.



Final model, validation vs. held-out test: ROC-AUC and F1 hold up (F1 even improves slightly), while PR-AUC and Precision@K drop toward the text-only baseline's level.

7. Discussion and Limitations

Does the image help? The honest answer is: modestly, and not conclusively. On validation data, across many iterations of tuning, the best multimodal model (exp07) beat the strongest text-only baseline by a sizeable margin. On a single, untouched test month, that margin nearly disappears on the primary metric (PR-AUC) while holding up on the secondary one (ROC-AUC). Given the well-understood mechanism (repeated validation-based model selection), we do not read this as “the image adds nothing” so much as “the image’s contribution is real but small, and was likely overstated by how many tuning decisions were routed through the same validation set.”

Why did the more sophisticated fusion approach fail? The controlled ablation in Section 6.3 gives a fairly definitive answer for this project’s specific implementation: it was not the *mechanism* (cross-attention vs. concatenation) but the *added trainable capacity and reduced training data* that hurt Model 1. A more thorough treatment might test cross-attention at *matched* capacity across a wider sweep, or attention over the full token sequence rather than pooled queries — both left unexplored given the project’s time budget.

Why did the Hybrid model succeed where SigLIP2 didn’t? Two compounding factors: a genuinely Hebrew-specific text encoder (AlephBERT) instead of a multilingual one only lightly exposed to Hebrew during pretraining, and much more conservative regularization discovered through iteration (fully-frozen vision backbone, higher weight decay, correctly-scaled early stopping). This suggests language-appropriateness of the text backbone mattered more, for this task and this language, than the theoretical advantage of SigLIP2’s jointly-pretrained, contrastively-aligned text/image embedding space.

Limitations. (1) A single train/val/test split (one val month, one test month) means metric volatility from month-to-month

variation is not separately quantified — a rolling-window or multiple-test-month evaluation would give a tighter estimate of the true PR-AUC gap. (2) Tags were excluded from the core model entirely (Section 3); their marginal value was never directly measured. (3) Thumbnail image quality varies substantially (Section 3) and, outside the controlled full-resolution filter used only for Model 1, was not otherwise adjusted for. (4) Only one random seed was used per configuration; results are not accompanied by variance estimates across seeds.

8. Conclusion

We built and rigorously compared six modeling approaches — four baselines isolating single modalities, staged SigLIP2 fine-tuning across three capacity regimes with two fusion mechanisms, and a Hebrew-specific hybrid architecture — to answer whether a news article’s thumbnail image predicts its relative popularity beyond its headline. The best model (AlephBERT + frozen ResNet50, tuned regularization) improved PR-AUC over a strong text-only baseline on validation data (0.240 vs. 0.212) but that advantage substantially narrowed on a genuinely held-out test month (0.212 vs. 0.212), while overall ranking quality (ROC-AUC) held up. We conclude the thumbnail carries a real but modest signal beyond the headline for this task, and that naive fusion of frozen multimodal embeddings is actively counterproductive — joint, task-specific fine-tuning with an appropriately-matched (here, Hebrew-specific) text encoder is what allowed any image contribution to surface at all.

References

1. M. Tschannen et al., “SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features,” arXiv:2502.14786, 2025.
2. F. S. Abousaleh, W.-H. Cheng, N.-H. Yu, and Y. Tsao, “Multimodal Deep Learning Framework for Image Popularity Prediction on Social Media,” IEEE Transactions on Cognitive and Developmental Systems, vol. 13, no. 3, 2021.